



DEPARTMENT OF  
COMPUTER SCIENCE



# SOFTWARE TESTING LIFE CYCLE AND DEBUGGING

**JEREMIAH O. BANDELE**

PhD Electrical/Electronic Engineering  
University of Nottingham



## LEARNING OBJECTIVES

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At the end of the lesson, you should be able to:

- Discuss software testing and why we need software testing for problem solving.
- Outline the software testing life cycle.
- Describe debugging and introduce some debugging tools.





## SOFTWARE TESTING

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Software testing can be explained as a process to find out if the output the computer is giving is the same as the expected output.

How can we tell if the computer is giving the right output or not? A functional test?





## SOFTWARE TESTING

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With software testing, we can locate errors in our program and also, confirm if there any missing requirements in the program.

The software testing process can either be carried out manually or with automated tools.



## **SOFTWARE TESTING LIFE CYCLE**

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We need a software testing life cycle so that while trying to test our programs, we follow well defined procedures to be sure that our software meets the set requirements and that it does not have defects. If there is no standard process like the software testing life cycle, which clearly defines the deliverables and objectives of each testing phase, we might not be sure that we are testing our software the right way.



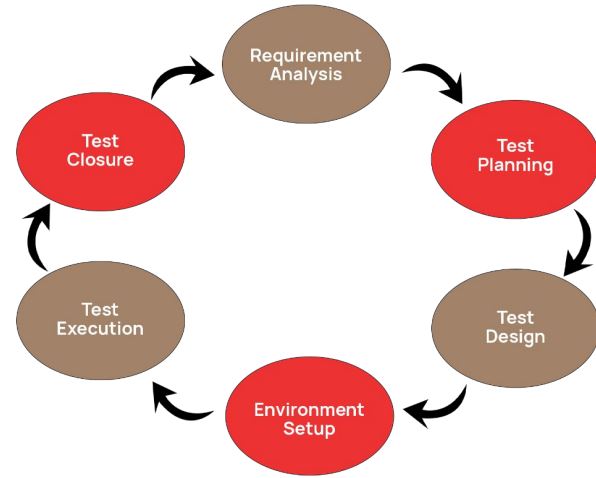
## SOFTWARE TESTING LIFE CYCLE: PHASES

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The software testing life cycle has six main phases.

These phases in sequential order are the:

- Requirement analysis phase
- Test planning phase
- Test case development phase
- Test environment setup phase
- Test execution phase
- Test cycle closure phase.





## **SOFTWARE TESTING LIFE CYCLE: PHASES**

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Each phase has entry and exit points.

The entry point is where the phase begins and the exit point is where the phase ends, and all the objectives of the phase have been achieved.



## REQUIREMENT PHASE

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This is the phase where we decide on what is to be tested.

The requirement phase includes activities such as a software requirement document review, stakeholders interview and the identification of the possible issues that might affect the software testing process.





## **TEST PLANNING PHASE**

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In this phase, the testing strategy, cost and duration are clearly defined.

The test planning phase includes activities such as the identification of testing objectives, testing environment, test cases, and test deliverables.

The assignment of roles to team members also takes place in this phase.



## **TEST CASE DEVELOPMENT PHASE**

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In this phase, the test cases are clearly defined. So, what is a test case?

A test case gives detailed step-by-step instructions on how a test should be performed. A test case specifies the test inputs, process and expected outputs.

The test case development phase includes activities such as the identification, writing, and review of test cases.



## **TEST ENVIRONMENT SETUP PHASE**

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In this phase, the software and hardware requirements for the test are specified. The test environment setup phase includes activities such as the test data and environment setup.



## TEST EXECUTION PHASE

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In this phase, the actual software testing takes place, using the test plans and cases developed in the earlier phases.

The test execution phase includes activities such as test implementation, retesting to fix defects, test results analysis, and reporting.



## **TEST CYCLE CLOSURE PHASE**

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In this final phase, checks are carried out to ensure that the testing objectives have been achieved and that the software is ready for use.

The test cycle closure phase includes activities such as defect identification and correction, summary report creation and feedback collection.



## DEBUGGING

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Bugs in our code can cause a software system to malfunction, and can lead to poor performance or incorrect results.

The process of identifying and resolving errors, or bugs, in a software system.

Debugging can be a time-consuming and complex task, but it is essential for ensuring that a software system is functioning correctly.





## **DEBUGGING: METHODS AND TECHNIQUES**

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Code Inspection: Manually reviewing the source code of a software system to identify potential bugs or errors.

Monitoring: This involves monitoring a software system for unusual behavior or performance issues that can indicate the presence of bugs or errors.

Logging: This involves recording events and messages related to the software system, which can be used to identify bugs or errors.



## **DEBUGGING: METHODS AND TECHNIQUES**

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Unit Testing: Involves testing independent units or components of a software system to identify bugs or errors.

Integration Testing: Involves testing the interactions between different components of a software system to identify bugs or errors.

System Testing: Involves testing the entire software system to identify bugs or errors.





## DEBUGGING: METHODS AND TECHNIQUES

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Debugging Tools: Various tools available for debugging includes debuggers, trace tools, and profilers that can be used to identify and resolve bugs.

Some of the widely used debuggers are;

- Radare2
- WinDbg
- Valgrind



## SUMMARY

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- Software testing is the process to find out if the output the computer is giving is the same as the expected out.
- The software testing life cycle has six main phases.Each phase has entry and exit points.
- Debugging is the process of identifying and resolving errors, or bugs, in a software system.





## FURTHER READING RESOURCES

- Murch, R. (2012). The Software Development Lifecycle- A Complete Guide. Richard Murch.
- Khannur, A. (2014). Structured software testing: the discipline of discovering. Partridge Publishing India
- Software Engineering I Debugging.  
<https://www.geeksforgeeks.org/software-engineering-debugging/>



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# THANK YOU

